

Assignment: Manual Clustering using K-Means and DBSCAN

Dataset (12 Points)

Point	X	Y
P1	1	2
P2	2	2
P3	2	3
P4	3	3
P5	8	7
P6	8	8
P7	7	7
P8	25	80
P9	24	78
P10	26	82
P11	50	50
P12	52	48

Part 1: K-Means Clustering (K = 3)

Tasks:

1. Assign points to nearest centroid using Euclidean distance.
2. Update centroids.
3. Repeat until convergence.
4. Provide final clusters and centroids.

Part 2: DBSCAN

Given $\epsilon = 3$ and MinPts = 2:

1. Compute neighborhood of each point.
2. Identify core, border, and noise points.
3. Form clusters using density reachability.
4. Provide final clusters and noise points.

Part 3: Comparison & Reflection

1. Compare K-Means vs DBSCAN results.
2. Identify how outliers are handled.
3. Which method performs better and why?